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OPEN ENDED LAB

Digital Logic Design

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INSTRUCTOR

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OEL statement

Simulate a circuit to perform Binary Addition using Full Adder IC. Get 4-bit Binary input A from the output of the 1st 4-bit Universal Shift Register IC performing Shift Right Operation and get 4-bit Binary input B from the output of the 2nd 4-bit Universal Shift Register IC performing Shift Left Operation. Hence, there will be four 4-bit Binary Addition.

- Show the Binary Adder Output (Sum) in Seven Segment Display.
- Show the Carry bit Output in the Red probe indicator.
- Draw a table to show binary outputs of the 1st 4-bit Universal Shift Register IC performing the Shift Right Operation
- Draw a table to show binary outputs of the 2nd 4-bit Universal Shift Register IC performing the Shift Left Operation.
- Draw a table to show binary outputs of Full Adder IC.
- Show your results in Multisim
- Submit Multisim file, Screenshots of the circuit, and a compiled report.

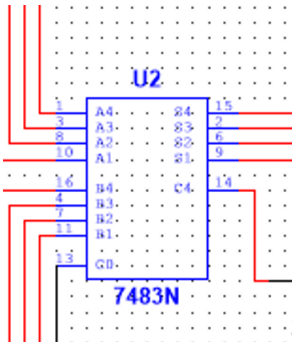
Components Used

- 4- Bit Full-Adder IC (74283) (x1)
- Universal Bidirectional Shift Register IC (74LS194D) (x2)
- BCD to Seven Segment Display Driver IC (74LS47D) (x1)
- 7-segment Display (Yellow, common Anode) (x1)
- Interactive Digital Constant (x16)
- Probes Digital (Green (x4), Blue (x4), Red (x1))
- Digital Power Vcc - 5V (x1)
- Ground (x1)

Schematic Capture:

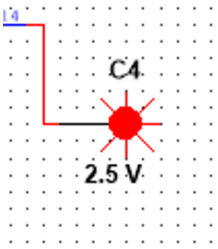
- Full-Adder IC

  Two binary values are needed for a full adder, along with a carry or overflow bit. A sum and an additional carry bit are the outputs. XOR, AND, and OR gates are the building blocks of full adders. To add bits to arbitrary lengths of bits, such as 32 or 64 bits, full adders are frequently coupled together.



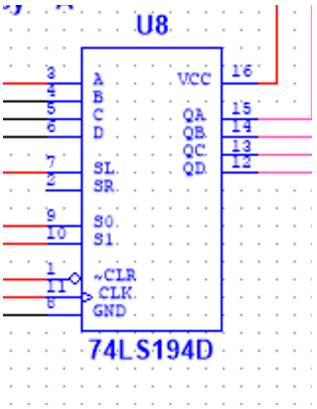
- Probe

  They are used to check the outputs of IC. Turn on at 1 and turn off at 0.



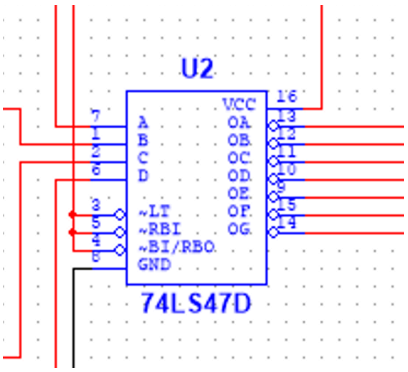
- Universal Shift Register

  A bidirectional shift register is capable of shifting in both directions. The Universal shift register is a combination design of a bidirectional shift register and a unidirectional shift register with parallel load provision.



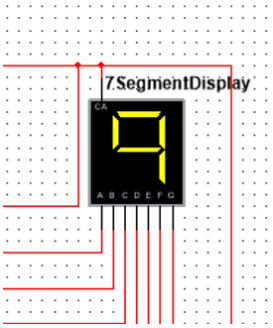
- Decoder

  IC number 7447 are circuits for BCD to seven segment converters and display from numbers 0 to 9. It decodes BCD in seven segmentations. Normally, IC7447 accepts input that is binary code values. The BCD is processed into seven segments shown as numbers 0 to 9 on a monitor.



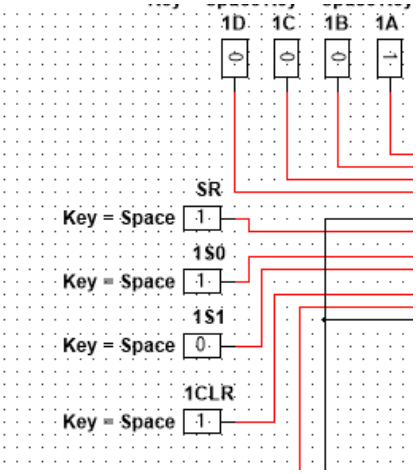
- 7-segment display

  A seven-segment display is a form of an electronic display device for displaying decimal numerals that is an alternative to the more complex dot matrix displays.



- Interactive Digital Constant

  This component is an interactive digital constant that produces a strong digital 1 (high) or 0 (low) state.



Truth Tables

Universal Shift Register:

S0	S1	Mode of Operation
0	0	Locked state (No change)
0	1	Shift-Left
1	0	Shift-Right
1	1	Parallel Loading

Table 3.1

Right Shifting							
Clock	**Q_A**	**Q_B**	**Q_C**	**Q_D**	**S_L**	**S_R**	**CLR**
1	1	0	0	0	X	1	1
2	1	1	0	0	X	1	1
3	1	1	1	0	X	1	1
4	1	1	1	1	X	1	1

Table 3.2

Left Shifting							
Clock	**Q_A**	**Q_B**	**Q_C**	**Q_D**	**S_L**	**S_R**	**CLR**
1	0	0	0	1	1	X	1
2	0	0	1	1	1	X	1
3	0	1	1	1	1	X	1
4	1	1	1	1	1	X	1

Table 3.3

Addition on first pulse				
$A= 0001_2$ $B=1000_2$				
A	**B**	**C_o**	**C_y**	**S**
1	0	0	0	1
0	0	0	0	0
0	0	0	0	0
0	1	0	0	1

Table 3.4

C_y S_4 S_3 S_2 S_1

$0\ 1\ 0\ 0\ 1_2 = 9_{10}$

Addition on second pulse				
$A= 0011_2$ $B=1100_2$				
A	**B**	**C_o**	**C_y**	**S**
1	0	0	0	1
1	0	0	0	1
0	1	0	0	1
0	1	0	0	1

Table 3.5

C_y S_4 S_3 S_2 S_1

$0\ 1\ 1\ 1\ 1_2 = 15_{10}$

Addition on third pulse				
$A= 0111_2$ $B=1110_2$				
A	**B**	**C_o**	**C_y**	**S**
1	0	0	0	1
1	1	0	1	0
1	1	1	1	1
0	1	1	1	0

Table 3.6

C_y S_4 S_3 S_2 S_1

$1\ \ 0\ 1\ 0\ 1_2 = 5_{10}$

Addition on fourth pulse				
$A= 1111_2$ $B=1111_2$				
A	**B**	**C_o**	**C_y**	**S**
1	1	0	1	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

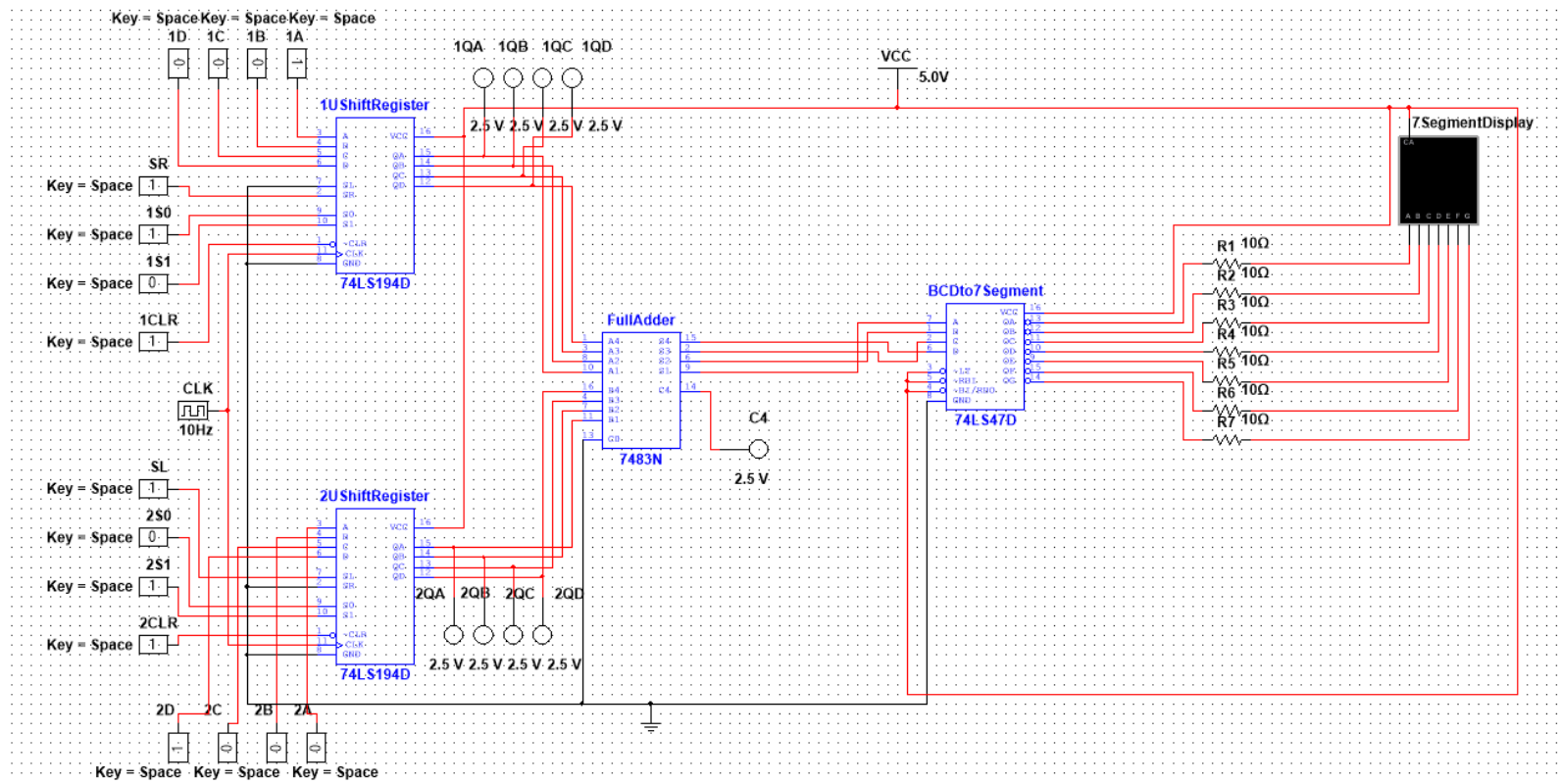
Table 3.7

C_y S_4 S_3 S_2 S_1

$1\ \ 1\ 1\ 1\ 0_2 = 14_{10}$

Seven Segment Display										
D	**C**	**B**	**A**	**a**	**b**	**c**	**d**	**e**	**f**	**g**
1	0	0	1	0	0	0	1	1	0	0
1	1	1	1	1	1	1	1	1	1	1
0	1	0	1	0	1	0	0	1	0	0
1	1	1	0	1	1	1	0	0	0	0

Complete Circuit

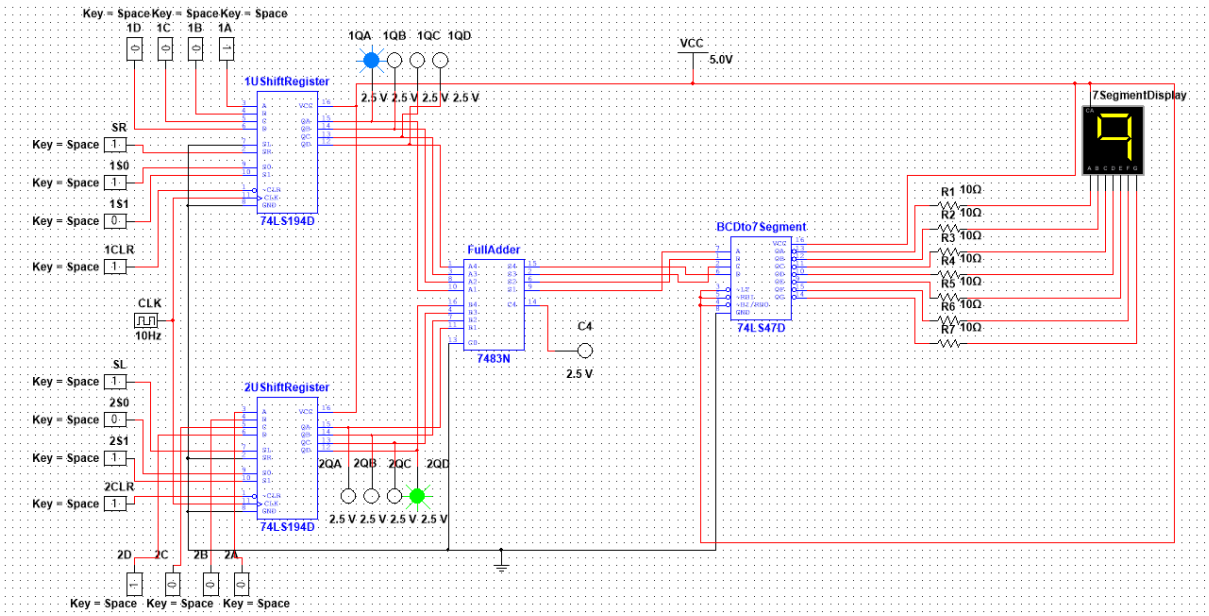


In the above circuit, two Universal Shift Registers are used because one input had to be shifted right and the second input had to be shifted left. To shift the bits right, mode controls S0 and S1 are given 1 and 0 logic respectively. To shift the bits left, mode controls S0 and S1 are given 0 and 1 logic respectively. The behaviour of Universal Shift Register can be seen in table 3.1, 3.2 and 3.3. The probes are used to display the shifting. After each shift, the Full Adder adds the two numbers and the decoder is used to convert BCD(binary coded decimal) to decimal so it can be displayed in Seven Segment Display. The display is shown in table 3.8.

Output

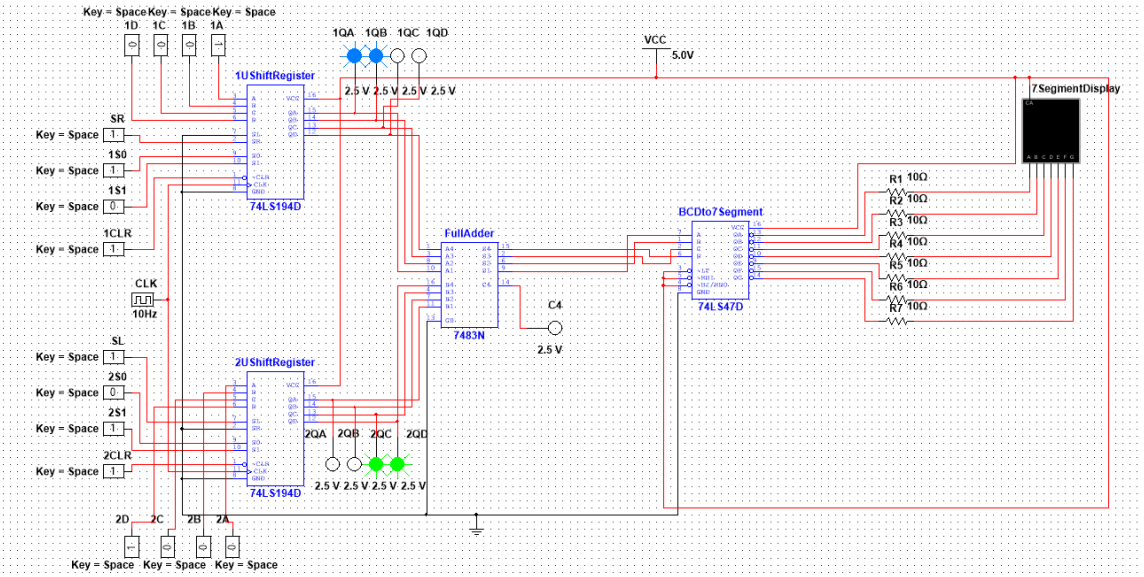
First Pulse:

At clock 1 the decimal value of right shift register is equal to 1 and the value of left shift register is equal to 8. Hence their addition must be equal to 9 after their binary values pass through the full adder IC showing 9 on the seven-segment display and carry out pin will be 0.



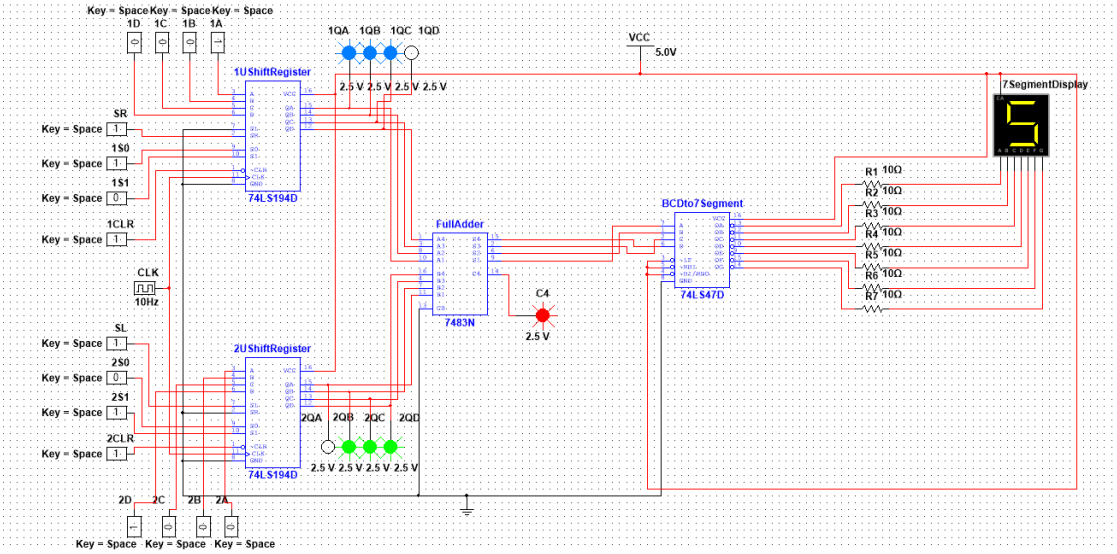
Second Pulse:

At clock 2 the decimal value of right shift register is equal to 3 and the value of left shift register is equal to 12. Hence their addition must be equal to 15 after their binary values pass through the full adder IC showing nothing on the seven-segment display and carry out pin will be zero.



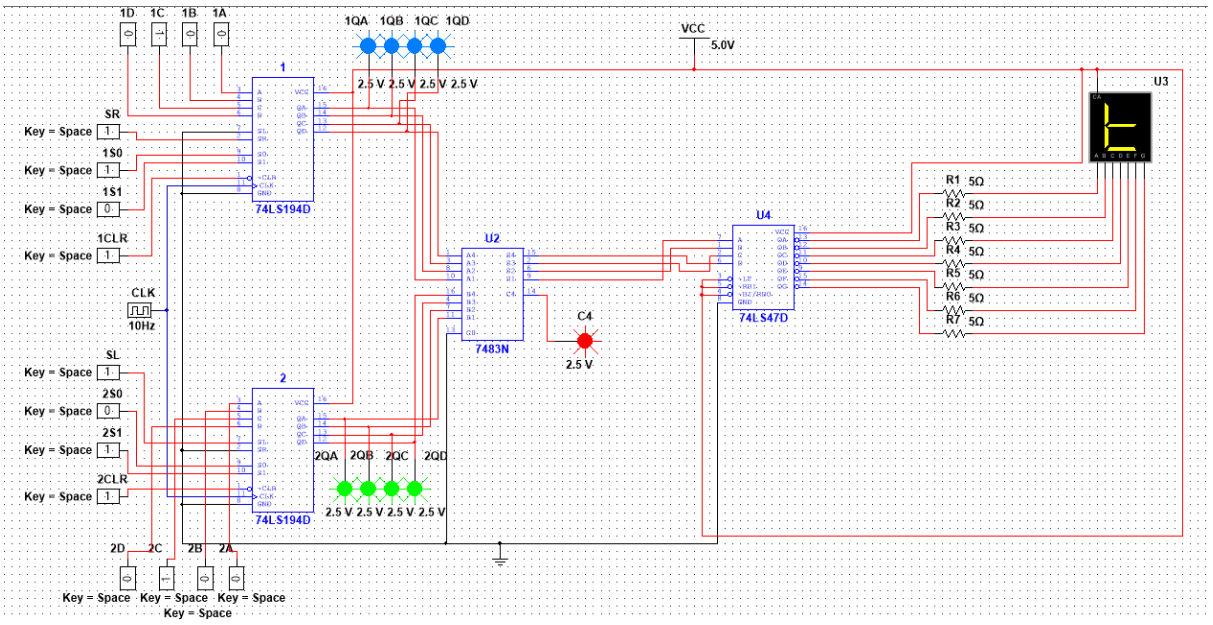
Third Pulse:

At clock 3 the decimal value of right shift register is equal to 7 and the value of left shift register is equal to 14. Hence their addition must be equal to 21 .After their binary values pass through the full adder IC the carry out pin will be high and the seven segment display will show 21-16=5.



Fourth Pulse:

At clock 4 the decimal value of right shift register is equal to 15 and the value of left shift register is equal to 15. Hence their addition must be equal to 30 .After their binary values pass through the full adder IC the carry out pin will be high and the seven-segment display will show 30-16=14.



Conclusion

When it comes to building this simulation, simplicity of use and effective management have been prioritised. Because of the independent switching and precise labelling, it is simple to use. All project requirements are being met and satisfied in full.